

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fourth Semester of B. Tech (IT) Examination****April 2018****IT245 Computer Organization & Microprocessor****Date: 30.04.2018, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the following questions.**

- a. If clock rate = 50 MHZ, find execution time for a program with 5,000 instruction, if the CPI for the program = 2.5 then what is the CPU (execution) time? If clock rate increases from 50 MHz to 1 GHz and the other factors remain the same, how many times faster will the computer be? Compute the execution time for both cases. **[03]**
- b. Two compilers being tested for 2.5 GHZ machine with 3 classes of instructions: **[04]**
- A (3 cycle), B (4 Cycles), and C (5 Cycles)
- Compiler 1: 5 A, 2 B, 3 C instructions
- Compiler 2: 10 A, 1 B, 4 C instructions
- Which code sequence executes the most instructions?
 - Which will be faster?
 - Calculate the CPI in both cases.

Q – 2.a Consider the following program: **[04]**

```

int va1=10;
char c[]="IT";
float var2;
int main()
{
    int *p;
    static int i=0;
    var2=20;
    p = (int *)malloc (sizeof (*p));
    *p = 99;
    add(var1, var2)
    return 0;
}

```

```

void add(int i, int j)
{
    int x = i;
    int y = j;
    int z = x + y;
}

```

Draw the memory image (text, data, bss, heap and stack segments) while processor is executing $\text{int } z = x + y$ statement.

Q – 2.b Write a short note on Average Memory Access Time (AMAT). **[04]**

Q – 2.c Consider a 32 M words main memory and 1024-block cache. Both partitioned into 32 words block. What is the size of additional memory for tags? Size of the cache? Consider direct mapping and 2-way set associative cache is used and word size 1 byte. Memory is byte addressable. **[06]**

OR

Q – 2.a Convert the following virtual addresses to physical address (Page Size = 4 KB). **[04]**

0x00000208 – 0x0006208, 0x00004120 - 0x0009120

0x0003
0x0004
0x0006
0x0008
0x0009
...
0x00f6

Consider the Virtual Address 0x00003208.

Find the cache line (index line) in virtually indexed physically tag.

Index line = 208

Also, mention the possible value of tag based on physical value calculated as the table given above.

Tag value = 0x0008

Q – 2.b What is Page? How paging is useful in the memory architecture? **[04]**

Q – 2.c A 32 KB 4 way set associative write-back cache is organized as multiple blocks, each of size 64-bytes. The processor generates 32-bit addresses. The cache controller maintains the tag information for each cache block comprising of the following. 1 Valid bit. What is the total size of memory needed at the cache controller to store meta-data (tags) for the cache? **[06]**

Q – 3.a Explain the hazards in the pipelining. Also, suggest the ways to overcome those hazards. **[06]**

- Q – 3.b** A 5-stage pipelined processor has Instruction Fetch (IF), Instruction Decode (ID), [08]
Operand Fetch (OF), Perform Operation (PO) and Write Operand (WO) stages. The
IF, ID, OF and WO stages take 1 clock cycle each for any instruction. The PO stage
takes 1 clock cycle for ADD and SUB instructions, 2 clock cycles for MUL instruction,
and 4 clock cycles for DIV instruction respectively. Operand forwarding is used in the
pipeline. What is the number of clock cycles needed to execute the following sequence
of instructions?

Instruction	Meaning of instruction
I0 :MUL R2 ,R0 ,R1	$R2 \leftarrow R0 * R1$
I1 :DIV R5 ,R3 ,R4	$R5 \leftarrow R3 / R4$
I2 :ADD R2 ,R5 ,R2	$R2 \leftarrow R5 + R2$
I3 :SUB R5 ,R2 ,R6	$R5 \leftarrow R2 - R6$
I4 :MUL R5 ,R5 ,R6	$R5 \leftarrow R2 - R6$

OR

- Q – 3.a** What is data forwarding in the pipelining? Why data forwarding cannot completely [06]
avoid stalling? Explain using suitable example.
- Q – 3.b** A 5 stage pipelined CPU has the following sequence of stages: [08]

IF — Instruction fetch from instruction memory,
RD — Instruction decode and register read,
EX — Execute: ALU operation for data and address computation,
MA — Data memory access - for write access, the register read at RD stage is used,
WB — Register write back.

Consider the following sequence of instructions:

I1 : L R0, 10c1; $R0 \leftarrow M[10c1]$
I2 : A R0, R0; $R0 \leftarrow R0 + R0$
I3 : A R2, R0; $R2 \leftarrow R2 - R0$

Let each stage take one clock cycle.

What is the number of clock cycles taken to complete the above sequence of
instructions starting from the fetch of I1?

SECTION – II

- Q - 4** Answer the following questions. [07]
- a. Why ARM processor architecture is considered the RISC architecture? [02]
- b. What is the 2's complement representation of (-96) in 8 bits? [02]
- c. Consider a processor is executing LDUR instruction. Fill the given table. [03]

Inst.	Reg2Loc	ALUSrc	MemtoReg	RegWrite	MemRead	Memwrite	Branch
LDUR							

- Q – 5.a** Explain all types of instructions format in the ARM processor. [07]
- Q – 5.b** Write the ARM V8 Assembly code for following code snippet. [07]

• While (i>3) { • a = a + 12345678 • b = b ^ 2; • i++; • }	• for(i=10; i > j; --i) { • k = k * 2 ; • i = i / 2 ; • }	• If (a == b) • b = c and 2 • else if (c == d) • d = NOT d
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OR

Q – 5.a Write ARMV8 Assembly code to create a function for finding factorial of a number. [07]

Q – 5.b Explain the following instructions using suitable example [07]

1) CBZ 2) CBNZ 3) BL

Q – 6.a Draw the data path of LDUR instruction. [04]

Q – 6.b Multiply 7 and -3 using booth's algorithm. Use 8-bit number representation. [05]

Q – 6.c i) What decimal value does the following 32-bit floating-point number represented in IEEE 754 Single Precision Floating Point represent? [05]

1 10000010 111101100000000000000000

ii) Also, state how to represent zero and infinity in IEEE Single Precision floating point.

OR

Q – 6.a Draw and explain the data path of STUR instruction. [04]

Q – 6.b Divide 7 and 3 using binary division algorithm. Use 8-bit number representation. [05]

Q – 6.c List the major IEEE 754 Rounding modes. [05]

Consider the floating-point numbers as given in the table. Tell their values after rounding to nearest even using G, R and S bits.

Sr No	S	E	Mantissa	G	R	S
1	1	10000000	1.1100000000000000000011111	0	0	0
2	0	10000010	0.011100000000000000000111	1	1	0
3	1	10000001	0.1110000000000000000001111	1	0	0
4	1	10000011	1.0010100000000000000001000	1	0	1
